

# TAREA T4: COMPLEMENTO DE UN NÚMERO-OPERACIÓN DE RESTO

## COMPLEMENTO AUTÉNTICO Y COMPLEMENTO RESTRINGIDO

T4: Alex Choque V.

C.I.: 13183020

I) Hallar el complemento CA y CR de los siguientes:

|                                                                                                                                                 |                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <p>a) <math>7434.0002_8</math></p> <p>Complemento Restringido</p> $\begin{array}{r} 7777.7777 \\ - 7434.0002 \\ \hline 0343.7775_8 \end{array}$ | <p>C.A.</p> $\begin{array}{r} 343.7775_8 \\ + 1 \\ \hline 343.7776_8 \end{array}$  |
| <p>b) <math>110010110_2</math></p> <p>C.R.</p> $\begin{array}{r} 111111111 \\ - 110010110 \\ \hline 001101001_2 \end{array}$                    | <p>C.A.</p> $\begin{array}{r} 1101001 \\ + 1 \\ \hline 1101010_2 \end{array}$      |
| <p>c) <math>39F4.06A_{16}</math></p> <p>CR<sub>16</sub></p> $\begin{array}{r} FFFFFFFF \\ - 39F4.06A \\ \hline C60B.F95_{16} \end{array}$       | <p>C.A.</p> $\begin{array}{r} C60B.F95 \\ + 1 \\ \hline C60B.F96_{16} \end{array}$ |
| <p>d) <math>3201.232_4</math></p> <p>C.R.</p> $\begin{array}{r} 3333.333 \\ - 3201.232 \\ \hline 132.101_4 \end{array}$                         | <p>C.A.</p> $\begin{array}{r} 132.101 \\ + 1 \\ \hline 132.102_4 \end{array}$      |
| <p>e) <math>3344.55_7</math></p> <p>C.R.</p> $\begin{array}{r} 6666.66 \\ - 3344.55 \\ \hline 3322.11_7 \end{array}$                            | <p>C.A.</p> $\begin{array}{r} 3322.11 \\ + 1 \\ \hline 3322.12_7 \end{array}$      |



$$\begin{array}{r} a) \quad 93A2.58_{12} \\ - \quad 67B.2_{12} \\ \hline \end{array}$$

$$\begin{array}{r} 93A2.58_{12} \\ - \quad 67B.2_{12} \\ \hline 8923.38_{12} \end{array}$$

C.R.

$$\begin{array}{r} + \quad BBBB.B \\ \quad 67B.2 \\ \hline + \quad BB540.90_{12} \\ \quad 93A2.58 \\ \hline + \quad 8923.28 \\ \quad \quad \quad C \\ \hline 8923.38_{12} \end{array}$$

C.A.

$$\begin{array}{r} + \quad BB540.90_1 \\ \hline + \quad BB540.90 \\ \quad 93A2.58 \\ \hline 8923.38_{12} \end{array}$$

$$\begin{array}{r} b) \quad 5532.35_8 \\ - \quad 64.723_8 \\ \hline \end{array}$$

$$\begin{array}{r} 5532.350 \\ - \quad 64.723 \\ \hline 5445.425 \end{array}$$

C.R.

$$\begin{array}{r} 7777777 \\ - \quad 64.723 \\ \hline + \quad 7713.054 \\ \quad 5532.350 \\ \hline + \quad 5445.424 \\ \quad \quad \quad 1 \\ \hline 5445.425_8 \end{array}$$

C.A.

$$\begin{array}{r} + \quad 7713.054_1 \\ \hline + \quad 7713.055 \\ \quad 5532.350 \\ \hline 5445.425_8 \end{array}$$

$$\begin{array}{r} c) \quad A036.F5_{16} \\ - \quad 94B3.4_{16} \\ \hline \end{array}$$

$$\begin{array}{r} A036.F5_{16} \\ - \quad 94B3.4_{16} \\ \hline 0B83.B5_{16} \end{array}$$

C.R.

$$\begin{array}{r} FFFFFFF \\ - \quad 94B3.4 \\ \hline + \quad F6B4C.B \\ \quad A036.F5 \\ \hline + \quad 00B83.A5 \\ \quad \quad \quad 1 \\ \hline B83.B5_{16} \end{array}$$

C.A.

$$\begin{array}{r} + \quad F6B4C.B_1 \\ \hline + \quad F6B4C.C \\ \quad A036.F5 \\ \hline 0B83.B5_{16} \end{array}$$



2) Calcular las operaciones de restas siguientes:

d) 
$$\begin{array}{r} 111000.0011_2 \\ - 100110.111_2 \\ \hline \end{array}$$

C.R.

C.F.

$$\begin{array}{r} 111000.0011_2 \\ - 100110.111_2 \\ \hline 100010101_2 \end{array}$$

$$\begin{array}{r} 111111.1111 \\ - 100110.1110 \\ \hline 011001.0001 \\ + 111000.0011 \\ \hline 010001.0100 \\ + 1 \\ \hline 10001.0101_2 \end{array}$$

$$\begin{array}{r} 1011001.0001 \\ + 1 \\ \hline 1011001.0010 \\ + 111000.0011 \\ \hline 10001.0101_2 \end{array}$$

e) 
$$\begin{array}{r} 21122.213 \\ - 00022.223 \\ \hline \end{array}$$

C.R.

C.A.

$$\begin{array}{r} 21122.213 \\ - 00022.223 \\ \hline 21022.223 \end{array}$$

$$\begin{array}{r} 22222.22 \\ - 00022.22 \\ \hline 22200.00 \\ + 21122.21 \\ \hline 21022.21 \\ + 1 \\ \hline 21022.223 \end{array}$$

$$\begin{array}{r} 22200.00 \\ + 1 \\ \hline 22200.01 \\ + 21122.21 \\ \hline 21022.223 \end{array}$$

f) 
$$\begin{array}{r} 51627.34_8 \\ - 6232.25_8 \\ \hline \end{array}$$

C.R.

C.A.

$$\begin{array}{r} 51627.34_8 \\ - 6232.25_8 \\ \hline 43375.07_8 \end{array}$$

$$\begin{array}{r} 77777.77 \\ - 6232.25 \\ \hline 71545.52 \\ + 51627.34 \\ \hline 43375.06 \\ + 1 \\ \hline 43375.07_8 \end{array}$$

$$\begin{array}{r} 71545.52 \\ + 1 \\ \hline 71545.53 \\ + 51627.34 \\ \hline 43375.07_8 \end{array}$$

$$\begin{array}{r} 9) \quad 483_{10} \\ - 2601_{10} \\ \hline \end{array}$$

$$\begin{array}{r} 483 \\ - 2601 \\ \hline - 2118 \end{array}$$

$$\begin{array}{r} - 10000 \\ - 2601 \\ \hline + 07399 \\ 483 \\ \hline 7882_{10} \end{array}$$

$$\begin{array}{r} 10000 \\ - 7882 \\ \hline = 2118_{10} \end{array}$$

$$\begin{array}{r} h) \quad 4235.21_6 \\ - 5201.3_6 \\ \hline \end{array}$$

$$\begin{array}{r} 4235.21_6 \\ - 5201.3_6 \\ \hline - 0522.05 \end{array}$$